MIL-C-50494 (MU)
22 February 1972

MILITARY SPECIFICATION

CAP BLASTING, PRACTICE M10
PARTS FOR, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts for and assembling and packing for one type of inert blasting cap designated as Cap, Blasting, Practice, M10.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Inspection Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1345

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DRAWINGS

U.S. ARMY MUNITIONS COMMAND

9251499 - Cap, Blasting, Practice M10

PUBLICATIONS

U.S. ARMY MUNITIONS COMMAND

ET-9251499 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Practice Blasting Cap.-The practice blasting cap shall comply with all requirements specified on Drawing (dwg.) 9251499, and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts and assemblies shall be fabricated in a thorough workmanlike manner. The parts and assemblies shall be free of cracks, chips, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.4 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 First Article Inspection

4.2.1 Prior to the start of regular production, the contractor shall submit a first article sample, as designated by the contract, for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of items which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing components shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and he shall submit a record of this inspection with the sample, including certificates of conformance of materials. A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit an additional sample or portion thereof of items for first article inspection. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.2.2 First Article Inspection Provisions

4.2.2.1 Blasting Cap Assemblies

4.2.2.1.1 Submission.-Thirty-two (32) complete assemblies shall be submitted to a Government approved laboratory designated by the contract.

4.2.2.1.2 Inspections to be performed.-The complete assemblies will be subjected to any or all of the inspections specified in 4.3.2.5, 4.3.2.6, 4.3.2.7 and 4.3.2.8 and the tests specified in 4.3.3 of this specification and any or all requirements of the applicable drawings.

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4.2.2.1.3 Rejection.-If any assembly fails to comply with the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection of the first article sample upon any failure of an item to comply with any of the requirements as specified.

4.3 Inspection Provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of complete assemblies shall contain parts of one kind from one supplier.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's

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designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and and AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraphs of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Rubber plug assembly (see dwg. 9251497 covering a detail of dwg. 9251499).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Length of bare wire under minimum	Gage	01001
Minor:	AQL 1.50 percent		
201.	Outside diameter of plug	Gage	01002
202.	Length of plug	Gage	01003
203.	Evidence of poor workmanship (see 3.3)	Visual	01004

4.3.2.2 Cup (see dwg. 8830953 covering a detail of dwg. 9251498).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.50 percent		
201.	Inside diameter, min	Gage	02001
202.	Outside diameter, max	Gage	02002
203.	Evidence of poor workmanship (see 3.3)	Visual	02003
204.	Length	Gage	02004

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4.3.2.3 Shorting prong (see dwg. 8861595 covering a detail of dwg. 8800910).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Length	Gage	03001
102.	Diameter of ends	Gage	03002
103.	Distance between centers of ends	Gage	03003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.3)	Visual	03004

4.3.2.4 Contact, female (see dwg. 8863763 covering a detail of dwg. 8800910).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Inside diameter of hole before pinch	Gage	04001
102.	Inside diameter of hole after pinch	Gage	04002
103.	Depth of hole before pinch	Gage	04003
104.	Depth of crimp hole	Gage	04004
Minor:	AQL 1.50 percent		
201.	Inside radius missing	Visual	04005
202.	Knurl incorrect or missing	Visual	04006
203.	Chamfer missing	Visual	04007
204.	Evidence of poor workmanship (see 3.3)	Visual	04008

4.3.2.5 Female connector with shorting plug (see dwg. 8800910 covering a detail of dwg. 9251498).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:		AQL 1.00 percent	
101.	Centerline distance between female contacts	Gage	05001
102.	Centerline distance between ends of shorting prong	Gage	05002
103.	Depth of top end of female contacts	Gage	05003
104.	Recessed depth to outer end of prong	Gage	05004
105.	Recessed depth of dust cap	Gage	05005
106.	Contour of connector, contact end to shoulder	Gage	05006
107.	Length of connector, contact end to shoulder	Gage	05007
108.	Presence of rubber or other foreign material in contact holes	Visual	05008
Minor:		AQL 1.50 percent	
201.	Bare wire showing between cable and female connector over max. prior to molding in connector	Gage	05009
202.	Incorrect trim at wire ends (prior to crimping in contact and molding in connector)	Gage	05010
203.	Evidence of poor workmanship (see 3.3)	Visual	05011

4.3.2.6 Practice blasting cap and connector assembly (see dwg. 9251498 covering a detail of dwg. 9251499).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Cup holes missing (see 6.4)	Visual	06001
Major:		AQL 0.65 percent	
101.	Length of cable	Gage	06002
102.	Shorting plug missing, damaged or loose	Visual-Manual	06003
103.	Connector damaged	Visual	06004
104.	Wire insulation damaged	Visual	06005

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Minor:	AQL 2.50 percent		
201.	Cable ends not tinned (prior to twisting)	Visual	06006
202.	Cable ends not twisted	Visual	06007
203.	Color of wire	Visual	06008
204.	Crimp incomplete	Visual	06009
205.	Evidence of poor workmanship (see 3.3)	Visual	06010

4.3.2.7 Spool (see dwg. 9227086 covering a detail of dwg. 9251499).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 2.50 percent		
201.	Length, max	Gage	07001
202.	Width, max	Gage	07002
203.	Height, max	Gage	07003
204.	Evidence of poor workmanship (see 3.3)	Visual	07004
205.	Color incorrect	Visual	07005

4.3.2.8 Cap, Blasting, Practice M10, (see dwg. 9251499).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Insulation of wire leads damaged	Visual	08001
Minor:	AQL 1.50 percent		
201.	Marking missing, incorrect or illegible	Visual	08002
202.	Wire protrudes beyond contour of spool (except at cutouts)	Visual	08003
203.	Evidence of poor workmanship (see 3.3)	Visual	08004

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4.3.2.9 Carton, sealed (see Section 5).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Evidence of loose contents	Manual	09001
102.	Carton damaged so that contents are exposed or liable to be exposed	Visual	09002
103.	Closing for shipment improper	Visual	09003
Minor:	AQL 0.65 percent		
201.	Marking misleading or improper	Visual	09004

4.3.3 Testing

4.3.3.1 Continuity and resistance test of assembly (see dwg. 9251499) - Major Defect - Code No. 10001.-This test shall be performed 100 percent. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4. Any assembly which fails to comply with the specified requirements shall be classed defective and removed from the lot.

4.3.4 Inspection equipment.-Equipment Tabulation Number ET-9251499 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement of parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.4 Test Methods and Procedures

4.4.1 Continuity and resistance test.-The shorting plug shall be removed and the resistance measured between female contacts and observations made for compliance with the specified requirement.

5. PREPARATION FOR DELIVERY

5. Packaging, packing and marking shall be in accordance with Level A or Level C requirements of MIL-STD-1169 as specified. Letter size and material requirements shall conform to paragraph 3.7.2.1 of MIL-STD-1169.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. The contractor is not required to furnish drawings of commercial inspection equipment, but a list of such equipment including manufacturer and catalog number shall be submitted for approval by the Government. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions.

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Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-M. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Critical defect inspection.-If the contractor is not manufacturing loaded M4 Blasting Cap Assemblies concurrently with manufacturing of Practice Caps, he may make written application to the Contracting Officer for authorization to delete the critical defect of paragraph 4.3.2.6 and substitute a major defect for inspection of holes on a sampling basis.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1345-A-296

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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